BIOL 343
Applied Bioinformatics

Counting

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Counting converts mapped reads to expression values

Expression values will be used by statistical models to identify DEGs

- Recall the goal of our RNA-seq experiments…
  - Treatment vs Control
  - Mutant vs Wild type
  - ***Identify differentially expressed genes (DEGs)***
- DEGs will be identified using statistical tests comparing ***expression values*** of transcripts/genes
- Expression values will be calculated based on the number of reads that ***align/map*** to a given genomic locus
- There are different ways to count reads

Counting converts mapped reads to expression values

Expression values will be used by statistical models to identify DEGs

Sample 1



9 reads per million

RNA

Library

Reads

Expression values

Differentially expressed gene?

Sample 2



18 reads per million

JBrowse

FILE ADD TOOLS HELP

New Session 10/26/2026, 8:23:42 AM

JBrowse

Zebrafish

NC_141034.1:11,961,401..11,965,102

3.7Kbp

Annotations

_015507215.1

_015507214.1

NM_001077607.1

The

Alignments RG:DMSO_1a 550

Alignments RG:DMSO_1b 800

Alignments RG:DMSO_1c 900

Alignments RG:DMSO_1d 600

Alignments RG:DMSO_1e 1600

Alignments RG:DMSO_2c 120

Alignments RG:DMSO_2d 200

Alignments RG:DMSO_3a 400

Alignments RG:DMSO_3c 550

Alignments RG:DMSO_3d 550

Alignments RG:cortisol_1a 80

Alignments RG:cortisol_2d 80

Alignments RG:cortisol_3a 220

Alignments RG:cortisol_3d 280

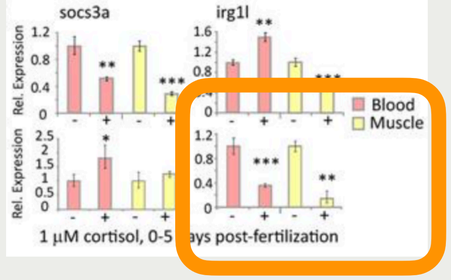
Alignments RG:cortisol_2c 110

Alignments RG:cortisol_2b 80

Alignments RG:cortisol_3b 140

Alignments RG:cortisol_2a 120

Higher expression in DMSO than cortisol…do we see that?



How are reads converted to expression values?

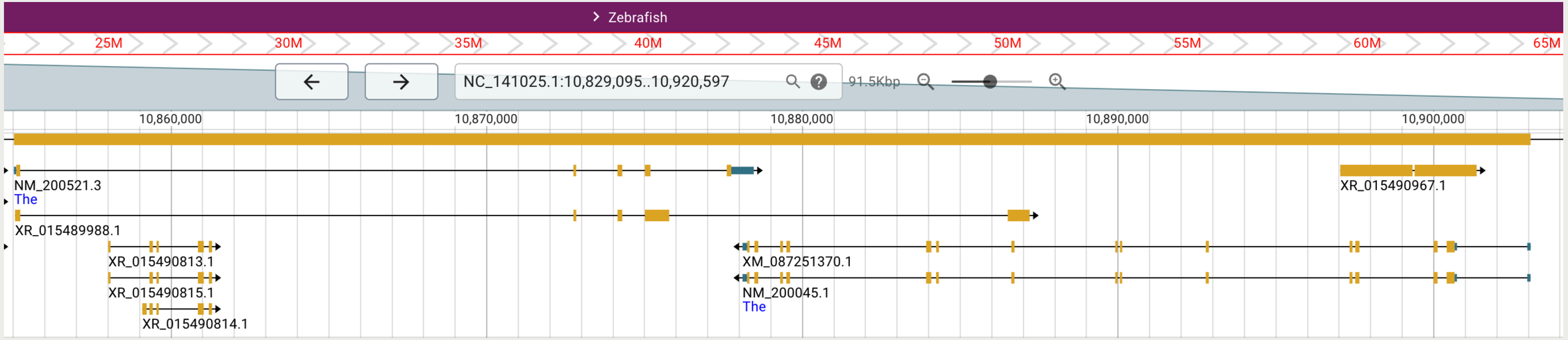
Many different situations must be considered

- What if a read only overlaps 1 nt of the gene?
- What if a read overlaps two genes?
- What if a gene has a lot of reads at one location and no reads at a different location?
- What if a read overlaps mostly intron?
- What if there are multiple isoforms? Do you count by transcript or by gene?
- How do you normalize for genes that have variable lengths?

The featureCounts approach

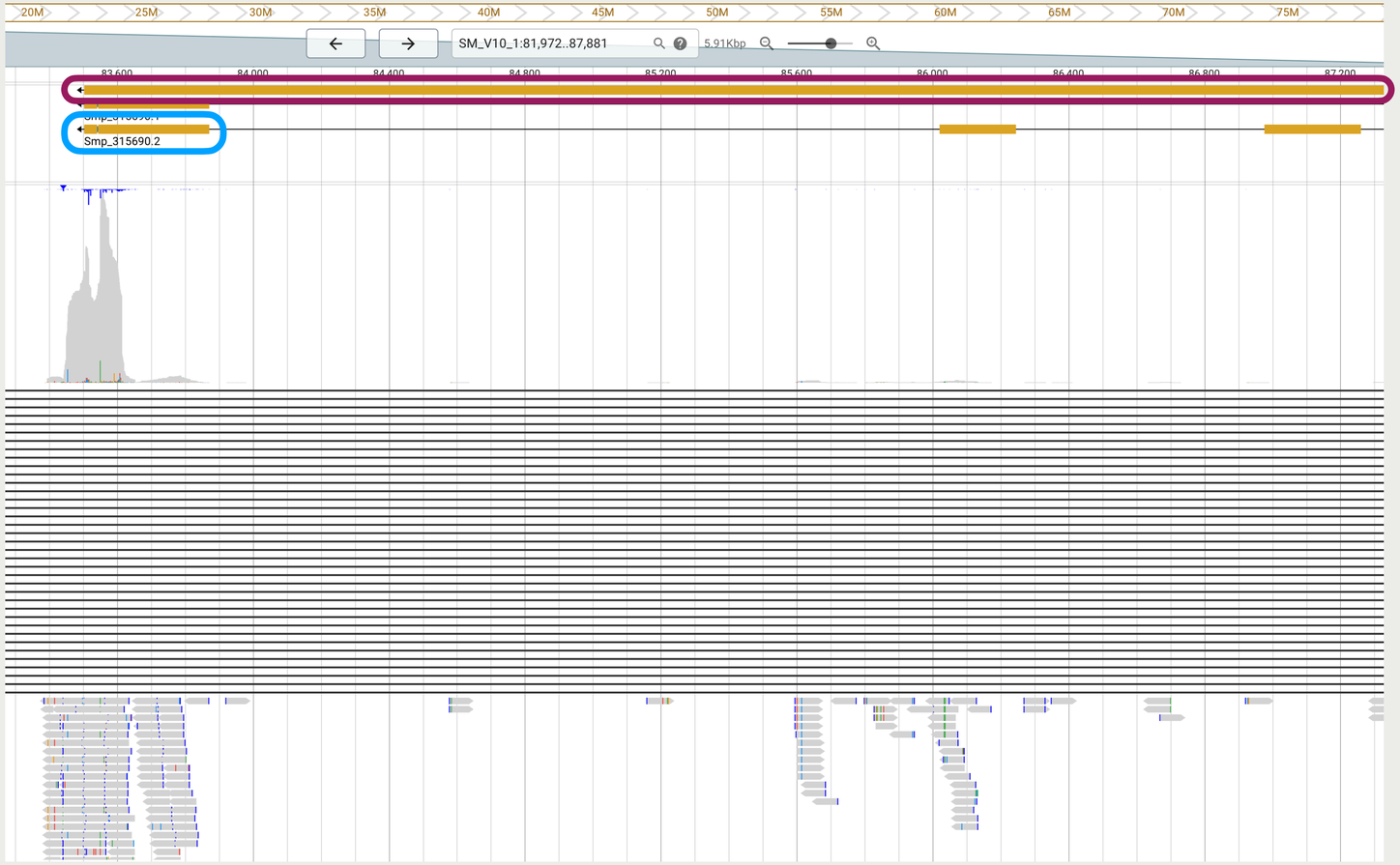
Features and meta-features

- Consider NC_141025.1:10,829,095..10,920,597
  - 4 total genes
  - Multiple isoforms of most genes
  - Genes on opposite strands overlapping



The featureCounts approach

Features and meta-features



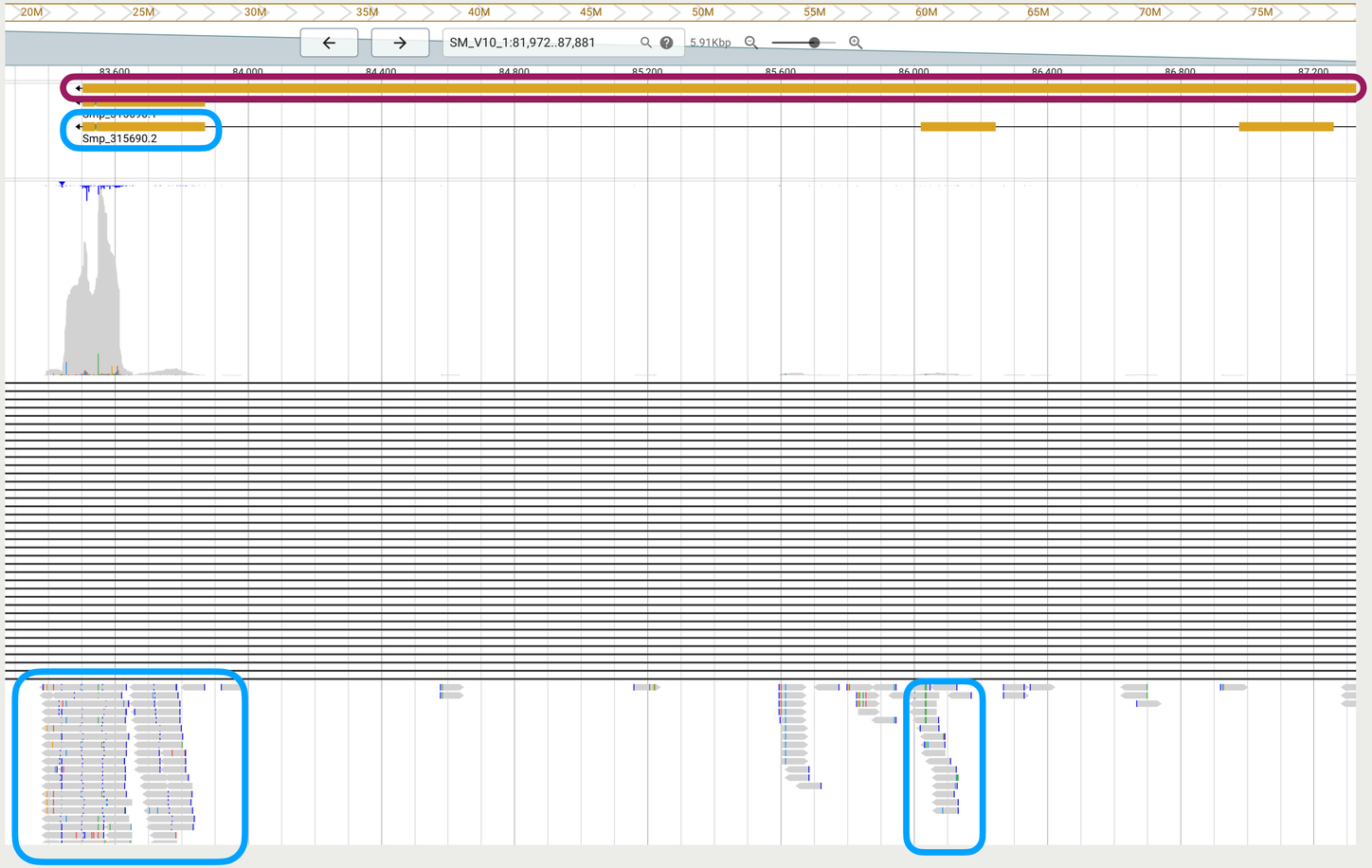
Feature = exon (by default)

Meta-feature = gene_id (by default)

- featureCounts first counts reads aligning to features (i.e., exons)
- Read is counted if it overlaps by 1 bp (by default)
- Then counts reads aligning to meta-features (i.e., genes)
  - Tallies all reads to an exon that is apart of that gene

The featureCounts approach

Features and meta-features



Hits to the feature

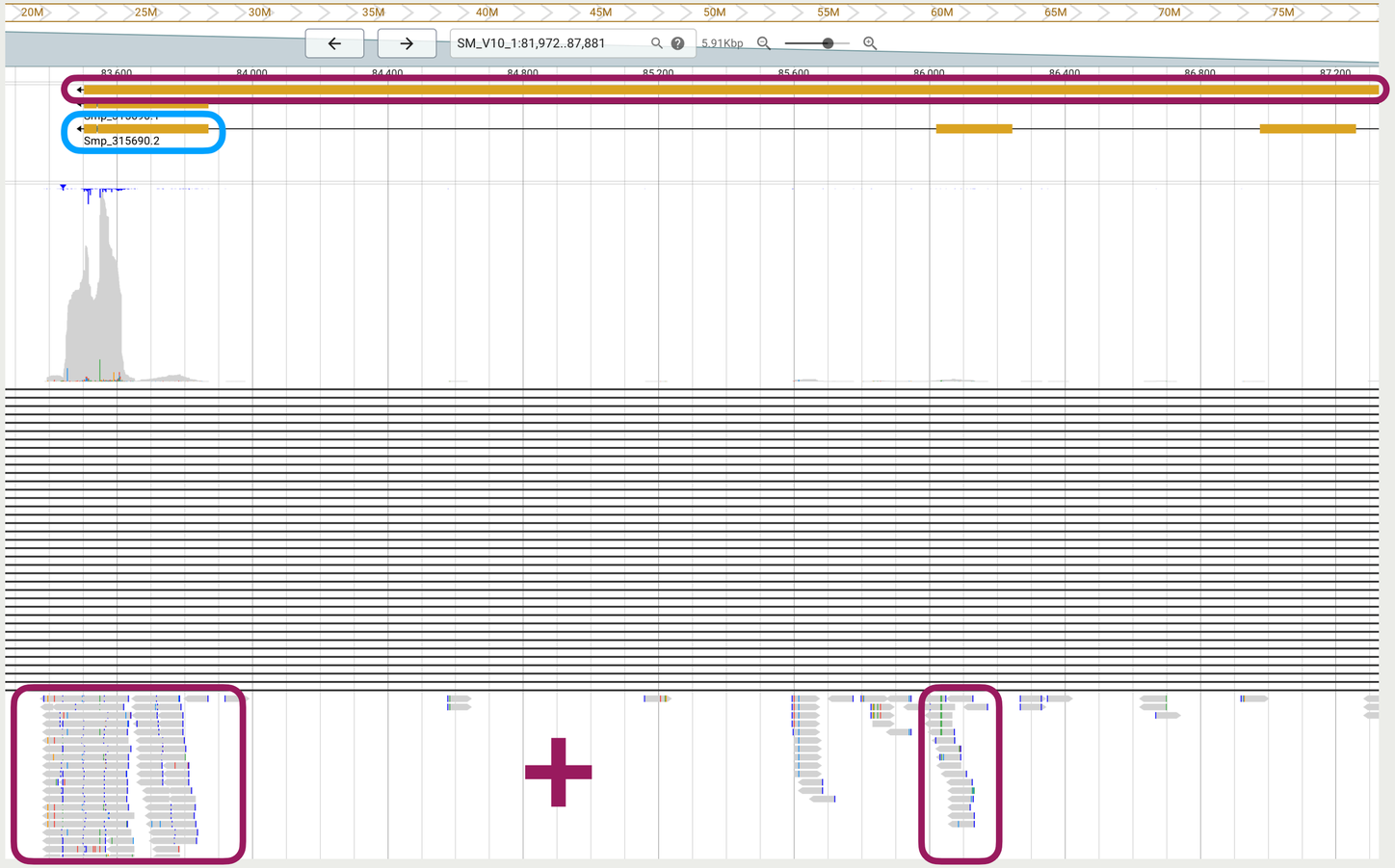
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The featureCounts approach

Features and meta-features



Hits to the meta-feature

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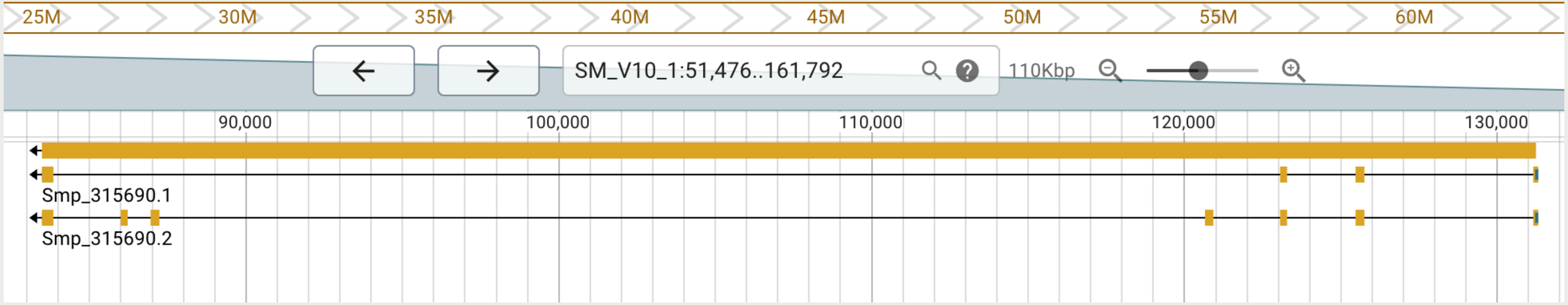
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The featureCounts approach

Features and meta-features

- featureCounts has excellent performance (written in C)
- Works well when transcript-level differential expression is not of interest
  - What about when you want to differentiate isoforms?
  - Only use hits that map unambiguously…



The featureCounts approach

Features and meta-features

- Output: integer counts at meta-feature level
- Matrix of integer values - the value in the *i*-th row and the *j*-th column represents the reads assigned to gene *i* in sample *j*
- Cannot do differential expression with only these counts
  - Unnormalized
    - Longer genes will necessarily have more reads
    - Counts will vary by library size (i.e., number of reads that aligned)
    - Differential expression tools will do the normalization for us